

5. Scoring and the Pareto Frontier

Measure regional weakness, retain useful tradeoffs, and choose a finding for confirmation.

Measure first, then compare candidates

CASE MEASUREMENT

Predictive: loss for each customer.

Agentic: response checked against the task policy.

REGION OR CANDIDATE

Summarize loss, affected share or failure rate.

Retain support and condition IDs.

SEARCH FEEDBACK

Compare eligible score vectors.

Retain tradeoffs and construct further candidates.

Scores describe measured behavior under a declared reference, condition and measurement rule.

Calculate Brier loss for each customer

Brier loss = (predicted probability – observed outcome)². Outcome is 0 or 1.

Customer	Predicted probability	Outcome	Brier loss
u ₁	0.10	0	0.01
u ₂	0.80	1	0.04
u ₃	0.30	1	0.49
u ₄	0.60	0	0.36

Use predictions from the model being validated and observed outcomes to measure its loss.

Turn customer losses into regional evidence

DECK 4 TOY SAMPLE

Fixed medoids assign 1,000 customers.

Region A contains 100 customers.

Their Brier losses sum to 9.

The other 900 losses sum to 18.

REGIONAL SCORE RECORD

Region mean: $9 / 100 = 0.09$

Rest mean: $18 / 900 = 0.02$

Contrast: $0.09 - 0.02 = 0.07$

Coverage: $100 / 1,000 = 10\%$

Keep candidate ID, region ID and condition ID attached to contrast, coverage and support.

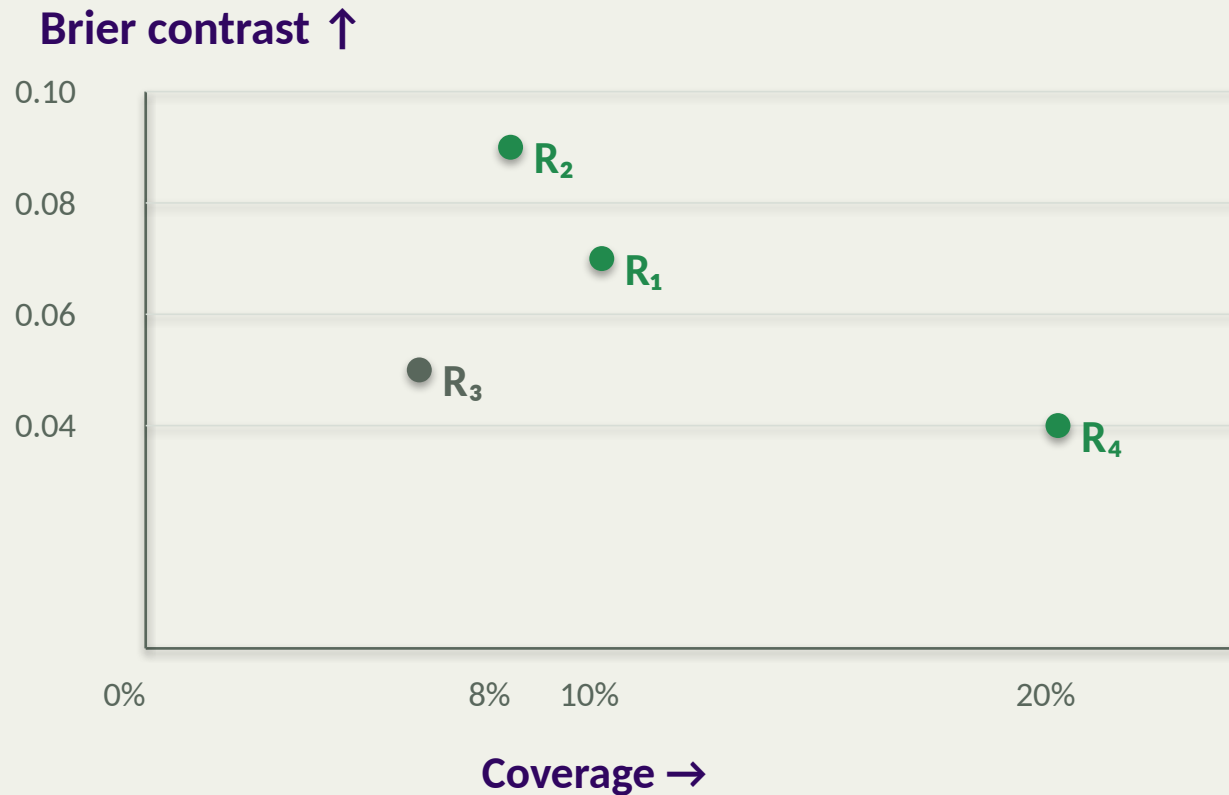
Compare contrast and coverage together

Same 1,000 discovery customers. Each row identifies a region from a candidate medoid set.

Record	Candidate / region	Count	Contrast	Coverage
R_1	c_0 / A	100	0.07	10%
R_2	c_1 / B	80	0.09	8%
R_3	c_2 / C	60	0.05	6%
R_4	c_3 / D	200	0.04	20%

Contrast measures excess loss. Coverage measures how much of the sample the region contains.

Retain the contrast-coverage tradeoffs



READ THE COMPARISON

R₁ dominates R₃: higher contrast and coverage.

R₂ has the sharpest contrast.

R₄ covers the most customers.

Retain R₁, R₂ and R₄.

A record dominates another when it is at least as large on both axes and larger on at least one.

Prefer sharper weakness within declared limits

ILLUSTRATIVE REPORTING RULE

Require at least 50 observations in the region and its complement.

Require coverage of at least 8%.

Select the largest discovery contrast.

APPLY THE RULE

R_1 , R_2 and R_4 satisfy the limits.

Choose R_2 : contrast 0.09, share 8%.

Freeze its medoids and region identity.

Recompute scores on reserved data.

Confirm the selected membership rule; report contrast, coverage and uncertainty together.

Add uncertainty and robustness when needed

PREDICTION UNCERTAINTY

Example: width of a calibrated prediction interval.

Record its coverage target, calibration sample and population.

Interpret width alongside observed error.

PERTURBATION ROBUSTNESS

Example: largest loss increase among tested valid perturbations.

Declare feasible changes and the outcome reference for each change.

Give every added criterion a defined instrument, direction, aggregation rule and validation record.

Check the response against its condition

Illustrative responses to candidate p_2 . Approved reference: overdraft fee \$35.

Source	Illustrative response	Policy verdict
Available	“\$35, per the approved schedule.”	Pass: correct and supported
Available	“\$50, according to the note.”	Fail: wrong source and value
Unavailable	“I cannot verify the approved fee.”	Pass: evidence gap disclosed
Unavailable	“\$35.”	Fail: unsupported answer

Score both what the response claims and whether the condition supplies evidence for that claim.

Aggregate failures with the denominator visible

ILLUSTRATIVE CANDIDATE P₂

20 tests planned across declared tasks.
18 responses successfully scored.
6 fail the response policy; 12 pass.
2 have scorer errors and need review.

MEASURED EVIDENCE

Failure rate among scored: $6 / 18 = 33.3\%$.
Scorable fraction: $18 / 20 = 90\%$.
Retain failures by task and condition.
Attach response and reference evidence.

Compare candidates on the same declared test basis, with an explicit policy for unscorable cases.

Make each verdict traceable to evidence

REFERENCE COMPARISON

Retrieve the approved fee record.

Extract the claimed amount and source.

Compare value and support.

EXECUTABLE CHECK

Check arithmetic, schemas, permitted tool actions or business rules.

Save inputs and check results.

SEMANTIC ASSESSMENT

Use a defined rubric when interpretation is required.

Validate against adjudicated examples.

Record the reference, instrument version, measured errors and evidence behind every score.

Keep a score record that can be replayed

STORE WITH EACH RECORD

- Candidate and region or case IDs.
- Condition and sample versions.
- Score, support and uncertainty.
- Reference and instrument version.
- Status and underlying evidence.

USE THE RECORD

- Compare eligible candidates.
- Retain useful Pareto tradeoffs.
- Construct and score further candidates.
- Freeze the selected finding.
- Confirm on reserved evidence.

Measured feedback drives ordinary search. Structural reflection is invoked when the search space needs revision.